

Word search puzzle (wordsearch)

In recent months, William has become passionate about *word search* puzzles. A word search puzzle consists of a grid of letters, in which there are one or more words to be searched in a vertical, horizontal or diagonal way.



Figure 1: A standard version of the puzzle.

William's version of this puzzle is a bit harder. There is *only one word* to be searched, but you are not limited to vertical/horizontal/diagonal words. You can compose a word in any way you like, starting from some letter and moving in any direction as long as you move to an *adjacent cell*: that means, you can move in 8 possible directions (two vertical, two horizontal and four diagonal). Note that William's rules do not forbid you to step on the same letter more than once: you have the freedom to compose the word even by reusing letters.

Compute how many times a given word appears in the word puzzle. This number can be quite large, so it's enough to print its value **modulo 1 000 000 007**.

🔔 The *modulo* operation ($a \bmod m$) can be written in C/C++ as `(a % m)` and in Pascal as `(a mod m)`. To avoid the *integer overflow* error, remember to reduce all partial results through the modulus, and not just the final result!

🔔 Among the attachments of this task you may find a template file `wordsearch.*` with a sample incomplete implementation.

Input

The first line contains the word S to be searched. The second line contains two integers H and W . The following H lines contain W letters each, and represent the letter grid forming William's word puzzle.



Output

You need to write a single line with an integer: how many times the given word appears in the letter grid (modulo 1 000 000 007).

Constraints

- $1 \leq H, W \leq 100$.
- The length of the word S does not exceed 1000.
- Every letter is alphabetical and lowercase (a to z).

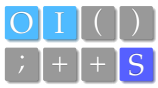
Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** [0 points]: Examples.
- **Subtask 2** [20 points]: $H = 1$: the input matrix is just one row.
- **Subtask 3** [20 points]: $H = W = 2$.
- **Subtask 4** [30 points]: $H, W \leq 10$.
- **Subtask 5** [30 points]: No additional limitations.

Examples

input.txt	output.txt
ciao 3 3 cia cio ciu	5
aaaaaaaaaaaa 5 8 aaaaaaaa aaaaaaaa aaaaaaaa aaaaaaaa aaaaaaaa	144253187



Explanation

In the **first sample case** there are 5 ways to compose the word “ciao”:

- Start from the first “c”, in the upper left corner, and go: right, right, down.
- Same starting point but go: down-right, up-right, down.
- Start from the second “c” and go: right, up-right, down.
- Same starting point but go: up-right, right, down.
- Start from the third “c” and go: up-right, up-right, down.